Compendium of Combinatorics

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Notation and Conventions. Let $\mathbb{N}$ denote the set of positive integers. For $n \in \mathbb{N}$, we define $[n] = \{1, 2, \ldots, n\}$. The cardinality of a set S is denoted by $|S|$. The binomial coefficient is denoted by $\binom{n}{k}$. The falling factorial is $(n)_k = n(n-1)\cdots(n-k+1)$. The natural logarithm is log, and e is the base of the natural logarithm.

Part I
Fundamental Counting Principles

1 The Basic Rules

- **Rule of Sum (Addition Principle):** If a task can be performed in m ways and another disjoint task in n ways, then there are $m + n$ ways to perform exactly one of the tasks. Formally, if $A \cap B = \emptyset$, $|A \cup B| = |A| + |B|$.
- **Rule of Product (Multiplication Principle):** If a task can be broken down into sequential stages, with n_1 ways to complete stage 1, n_2 ways for stage 2, etc., then the total number of ways is $\prod_{i=1}^{k} n_i$.
- **Inclusion-Exclusion Principle (PIE):** For finite sets $A_1, \ldots, A_n$:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i} |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n-1} |A_1 \cap \cdots \cap A_n|$$

- **Complementary Counting:** $|A| = |U| - |A^c|$, where U is the universal set.

2 Permutations and Combinations

- **Permutations (Order Matters):** The number of ways to arrange k distinct objects chosen from n distinct objects is:

$$P(n, k) = (n)_k = \frac{n!}{(n-k)!}$$

- **Combinations (Order Does Not Matter):** The number of ways to choose a subset of k objects from n distinct objects is:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- **Permutations with Repetition:** The number of strings of length k from an alphabet of size n is n^k.
- **Permutations of a Multiset:** The number of distinct arrangements of n objects where there are n_1 objects of type 1, n_2 of type 2, etc., is the multinomial coefficient:

$$\binom{n}{n_1, n_2, \ldots, n_k} = \frac{n!}{n_1! n_2! \ldots n_k!}$$

Part II
Binomial Coefficients and Identities

3 Core Identities

- **Symmetry:** $\binom{n}{k} = \binom{n}{n-k}$.
- **Pascal's Rule:** $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.
- **Absorption Identity:** $k\binom{n}{k} = n\binom{n-1}{k-1}$.
- **Vandermonde's Identity:** $\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k}$.
- **Hockey-Stick Identity:** $\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}$.

4 The Binomial Theorem

For any $x, y \in \mathbb{R}$ and $n \in \mathbb{N}$:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

- **Corollary 1:** $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.
- **Corollary 2:** $\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$ for $n \geq 1$.

5 The Multinomial Theorem

$$(x_1 + x_2 + \cdots + x_m)^n = \sum_{k_1 + \cdots + k_m = n} \binom{n}{k_1, \ldots, k_m} x_1^{k_1} \ldots x_m^{k_m}$$

Part III
Stars and Bars (Distributions)

The number of ways to distribute n identical items into k distinct bins:

1. **No restrictions (non-negative solutions):**

$$\binom{n+k-1}{k-1}$$

2. **Each bin gets at least one item (positive solutions):**

$$\binom{n-1}{k-1}$$

3. **Each bin gets at least r items:**

$$\binom{n-rk+k-1}{k-1}$$

Part IV
Recurrence Relations & Generating Functions

6 CommonRecurrences

- **Fibonacci Sequence:** $F_n = F_{n-1} + F_{n-2}$, with $F_0 = 0, F_1 = 1$. Closed form (Binet's Formula): $F_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$.
- **Catalan Numbers:** $C_n = \frac{1}{n+1}\binom{2n}{n}$. They satisfy $C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$. They count binary trees, valid parenthesis sequences, and polygon triangulations.
- **Derangements (D_n):** Permutations with no fixed points. $D_n = n!\sum_{k=0}^{n} \frac{(-1)^k}{k!} \approx \frac{n!}{e}$.

7 Generating Functions

- **Ordinary Generating Function (OGF):** $G(x) = \sum_{n=0}^{\infty} a_n x^n$.
  - Used for sequences where order doesn't matter (combinations).
  - *Example:* For $a_n = 1$, $G(x) = \frac{1}{1-x}$.
- **Exponential Generating Function (EGF):** $E(x) = \sum_{n=0}^{\infty} a_n \frac{x^n}{n!}$.
  - Used for sequences where order matters (permutations).
  - *Example:* For $a_n = n!$, $E(x) = \frac{1}{1-x}$.

Part V
Pigeonhole Principle & Extremal Combinatorics

8 Pigeonhole Principle

- **Basic:** If n items are placed into m containers and $n > m$, at least one container must contain at least $\lceil n/m \rceil$ items.
- **Generalized:** If N objects are placed into k boxes, at least one box contains $\lceil N/k \rceil$ objects.
- **Erdős–Szekeres Theorem:** Any sequence of $n^2 + 1$ distinct real numbers contains a monotonic subsequence of length $n + 1$.

9 Ramsey Theory

- **Ramsey's Theorem:** For any $r, s \geq 1$, there exists a minimum integer $R(r, s)$ such that any 2-coloring of the edges of the complete graph $K_{R(r,s)}$ contains a red K_r or a blue K_s.
- **Known Values:** $R(3, 3) = 6$, $R(3, 4) = 9$, $R(4, 4) = 18$.

10 Extremal Graph Theory

- **Turán's Theorem:** The maximum number of edges in an n-vertex graph that does not contain a clique of size $r+1$ is given by the Turán graph $T(n, r)$, with exactly $\left(1 - \frac{1}{r}\right) \frac{n^2}{2} + O(n)$ edges.

Part VI
The Probabilistic Method

A non-constructive technique to prove the existence of a combinatorial object by showing it has a positive probability of being generated randomly.

- **Union Bound:** $\mathbb{P}(\bigcup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} \mathbb{P}(A_i)$.
  - *Use Case:* If the sum of probabilities of "bad events" is strictly less than 1, then there exists an outcome where no bad events occur.
- **Linearity of Expectation:** $\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$. (Does not require independence).
  - *Use Case:* If $\mathbb{E}[X] = \mu$, then there must exist an outcome where $X \geq \mu$ and an outcome where $X \leq \mu$.

- **Lovász Local Lemma (LLL):** Let $A_1, \ldots, A_n$ be "bad events" such that each A_i is mutually independent of all but at most d other events. If $\mathbb{P}(A_i) \leq p$ for all i and $e \cdot p \cdot (d+1) \leq 1$ (where e is the base of the natural log), then $\mathbb{P}(\bigcap \overline{A_i}) > 0$.

Part VII
Discrepancy Theory

- **Standard Definition:** For a set system $\mathcal{F}$ on a ground set V, the discrepancy is $\text{disc}(\mathcal{F}) = \min_{\chi: V \to \{-1,1\}} \max_{S \in \mathcal{F}} |\sum_{x \in S} \chi(x)|$.
- **Matrix Form:** Minimizing $\|A\mathbf{x}\|_\infty$ where A is the incidence matrix and $\mathbf{x} \in \{\pm 1\}^n$.
- **Spencer's Theorem (Six Standard Deviations):** For any set system with n elements and m sets, there exists a coloring with discrepancy at most $O(\sqrt{m \log(n/m)})$. If $m = n$, the discrepancy is $O(\sqrt{n})$.

Part VIII
Asymptotics & Stirling's Approximation

- **Stirling's Approximation:**
$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$
Equivalently:
$$\log(n!) = n \log n - n + O(\log n)$$
- **Approximating Binomial Coefficients:**
$$\binom{n}{k} \approx \frac{n^n}{k^k (n-k)^{n-k}} \sqrt{\frac{n}{2\pi k(n-k)}}$$
- **Entropy Bound:** For $0 < p < 1$, where $k = pn$:
$$\binom{n}{pn} \approx 2^{nH(p)}$$
where $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ is the binary entropy function.